

Masked Diffusion Language Models for Structured-to-Text Generation: A Comparative Study on the ToTTo Dataset

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May 10, 2026

Abstract

Autoregressive language models such as GPT-2 have long dominated structured-to-text generation tasks, producing fluent descriptions from tabular data through left-to-right token generation. However, their sequential generation process may not be ideally suited for inputs such as tables that inherently lack a linear order. In this work, we investigate whether Masked Diffusion Language Models (MDLMs), which generate text through iterative bidirectional denoising, can offer advantages over autoregressive models on the table-to-text task.

We evaluate both GPT-2 and a pretrained MDLM (Qwen3-0.6B-diffusion-mdlm-v0.1) on the ToTTo benchmark in zero-shot and few-shot settings, measuring performance via BERTScore, BLEU, ROUGE-L, and Self-BLEU. Our results reveal that while GPT-2 benefits substantially from few-shot prompting, the MDLM achieves higher BERTScore in zero-shot conditions but shows a notable drop in few-shot settings, suggesting current limitations in leveraging in-context examples.

1. Introduction

The automatic generation of natural language descriptions from structured data — known as *structured-to-text* or *table-to-text* generation — is a fundamental challenge in Natural Language Processing (NLP). Applications span automatic report generation, data summarization, and question-answering from knowledge bases.

Most existing systems rely on autoregressive (AR) language models such as GPT-2 or T5, which generate text sequentially from left to right. While highly fluent, these models impose a linear generation process on inherently non-linear structures such as tables.

Tables do not possess a canonical reading order. Information can be interpreted row-wise, column-wise, or through highlighted cells. Consequently, a generation model capable of bidirectional reasoning may better capture the relational structure of tabular inputs.

Masked Diffusion Language Models (MDLMs) generate text through iterative denoising rather than sequential prediction. Starting from a fully masked sequence, the model progressively predicts tokens using bidirectional attention over the entire output space. This property makes MDLMs theoretically appealing for structured conditional generation.

In this paper, we compare GPT-2 and a pretrained MDLM on the ToTTo benchmark in zero-shot and few-shot settings without fine-tuning.

Our contributions are:

1. A comparative evaluation of autoregressive and diffusion-based language models on table-to-text

generation.

2. An analysis of zero-shot and few-shot prompting for both paradigms.
3. An evaluation using semantic, lexical, and diversity metrics.

2. Dataset and Task Description

We use the **ToTTo** dataset, a controlled table-to-text benchmark constructed from Wikipedia tables. Each example contains:

- A Wikipedia table,
- Highlighted cells,
- One or more reference descriptions.

The references are manually curated to ensure factual correctness and unambiguity, making ToTTo particularly suitable for evaluating factual grounding.

The original training set contains approximately 120,000 examples, while the development set contains around 7,700 examples.

Due to computational limitations of a Google Colab T4 environment, we use:

- 5,000 training examples,
- 500 evaluation examples.

For qualitative MDLM analysis, evaluation is restricted to the first 50 examples.

3. Data Preprocessing

Since both GPT-2 and MDLM operate on textual inputs, the structured tabular data from ToTTo was transformed into a linearized textual representation suitable for conditional generation.

For each example, only the highlighted cells were retained in order to focus the generation process on the most relevant information and reduce irrelevant contextual noise. Each highlighted cell was serialized using a structured `column : value` format, preserving both the semantic role of the column and the associated content. Multiple highlighted cells were concatenated into a single textual sequence separated by delimiters.

An example of serialized input is shown below:

```
Country : France | Capital : Paris  
| Population : 2.1 million
```

For the few-shot setting, several demonstration examples were concatenated before the query example using an in-context prompting strategy. Each demonstration consisted of a serialized table followed by its corresponding reference description. In the zero-shot setting, only the serialized input table was provided to the model.

Finally, all textual inputs were tokenized using the tokenizer associated with each respective model. Padding and truncation were applied when necessary to ensure compatibility with the maximum context length supported by the models. Generated descriptions were limited to 64 tokens to reduce inference cost and stabilize generation.

4. Models

4.1. GPT-2

GPT-2 GPT-2 small (117M parameters) serves as the autoregressive baseline for this comparative study. The model relies on a unidirectional transformer architecture to generate text sequentially, token-by-token, from left to right. For the purposes of this evaluation on the ToTTo dataset, the GPT-2 model itself was not fine-tuned; instead, the evaluation of its generation capabilities was conducted entirely through in-context prompting.

4.2. MDLM

MDLM For the diffusion-based approach, the study evaluates Qwen3-0.6B-diffusion-mdlm-v0.1, a masked diffusion language model adapted and fine-tuned from the base autoregressive Qwen3-0.6B architecture. This model was implemented and trained using the open-source **dLLM** (Simple Diffusion Language Modeling) framework. Unlike autoregressive models, a base Masked Diffusion Language Model (MDLM) generates text through an iterative, bidirectional denoising process rather than sequential left-to-right prediction. The generative inference begins with a sequence composed entirely of

masked tokens (an absorbing state), which the model progressively replaces with actual tokens in a random order

Mathematically, the forward masking process marginalizes to a categorical distribution that interpolates between the clean data x . Mathematically, the forward masking process marginalizes to a categorical distribution that interpolates between the clean data x and the absorbing masked state m :

$$q(z_t|x) = \text{Cat}(z_t; \alpha_t x + (1 - \alpha_t)m)$$

where α_t strictly decreases over time. To train the reverse denoising process, the base MDLM optimizes a continuous-time, Rao-Blackwellized variational lower bound (NELBO), which elegantly simplifies to a weighted average of masked language modeling (MLM) losses. For a sequence of length L , the objective is defined as:

$$\mathcal{L}_{NELBO}^{\infty} = \mathbb{E}_q \int_{t=0}^{t=1} \frac{\alpha'_t}{1 - \alpha_t} \sum_{\ell} \log \langle x_{\theta}^{\ell}(z_{1:L}^t, t), x^{\ell} \rangle dt$$

The efficiency of this model heavily relies on the "SUBS" parameterization during the reverse diffusion phase, which enforces two fundamental rules: Zero Masking Probabilities (the probability of generating a mask token is mathematically reduced to zero) and Carry-Over Unmasking (any unmasked token from a previous step is carried over and remains unchanged in subsequent steps). This allows the network to focus exclusively on predicting the missing words using bidirectional attention.

5. Evaluation Metrics

We evaluate models using:

- **BERTScore**: semantic similarity,
- **BLEU**: lexical overlap,
- **ROUGE-L**: longest common subsequence recall,
- **Self-BLEU**: diversity across generations.

6. Results and Discussion

6.1. Quantitative Comparison

The evaluation results obtained with the GPT-2 and MDLM models are illustrated in Figures 1 and 2, respectively.

BERTScore BERTScore measures the semantic similarity between the generated prediction and the ground-truth reference. Higher scores indicate stronger semantic alignment and therefore better generation quality. In both the few-shot and zero-shot settings, the MDLM model achieves the highest scores, with values of 0.5969 and 0.5636, respectively. In the zero-shot setting, the performance gap between the two models is substantial, with GPT-2 scoring approximately 0.21 points lower (0.3886) than MDLM.

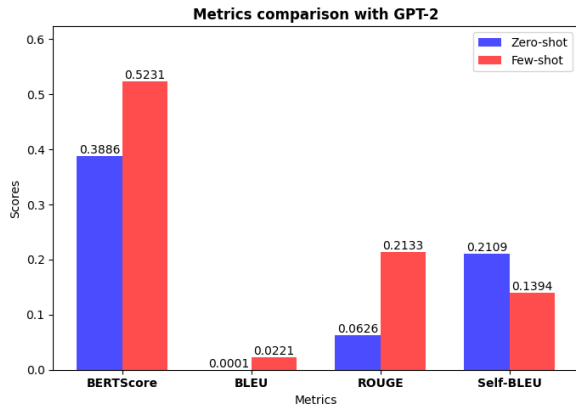


Figure 1: Zero-Shot vs. Few-Shot Performance of GPT-2

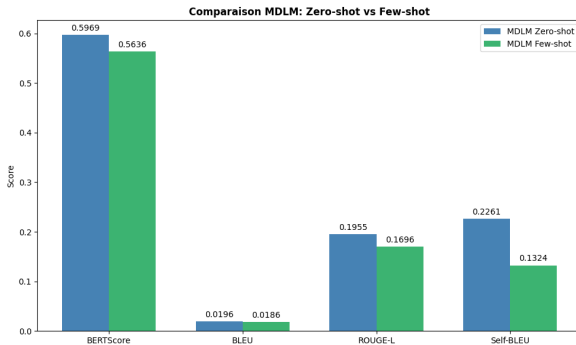


Figure 2: Zero-Shot vs. Few-Shot Performance of MDLM

In contrast, the difference is less pronounced in the few-shot setting, where GPT-2 remains only about 0.04 (0.5231) points below MDLM.

BLEU The BLEU (Bilingual Evaluation Understudy) score evaluates the similarity of n -grams between a generated text and a reference text. A higher BLEU score indicates that the generated text is lexically closer to the reference, whereas a lower score reflects weaker lexical similarity. In the zero-shot setting, MDLM achieves a higher BLEU score (0.0196) than GPT-2 (0.0001), indicating better lexical alignment with the reference texts. However, in the few-shot setting, the trend is reversed: GPT-2 obtains a superior BLEU score (0.0221) compared to MDLM (0.0186), suggesting that GPT-2 captures lexical similarities more effectively in this configuration.

ROUGE-L The ROUGE-L metric (Recall-Oriented Understudy for Gisting Evaluation – Longest Common Subsequence) evaluates the quality of a generated text by measuring the longest common subsequence (LCS) between the prediction and the reference text. A higher ROUGE-L score indicates that the generated text shares a greater degree of structural and content overlap with the reference, while preserving word order. Similarly to BLEU, in the zero-shot setting MDLM outperforms GPT-2, achieving a score of 0.1955 compared to 0.0626 for GPT-2. However, in the few-shot setting, GPT-2 performs better, reaching 0.2133 versus 0.1696 for MDLM.

Self-BLEU Self-BLEU is a metric that measures how similar generated texts are to each other by using BLEU as a similarity measure. More specifically, it evaluates the diversity of a set of generated texts by computing the BLEU score of each sample against the rest of the generated outputs. A higher Self-BLEU score indicates lower diversity, meaning that the generated texts are very similar to each other. Conversely, a lower score reflects higher diversity and greater variation among the generated samples. In both the zero-shot and few-shot settings, the results are broadly similar for the two models. However, a slight advantage can be observed for GPT-2 in the zero-shot setting (GPT-2: 0.2109 vs MDLM: 0.2261), indicating marginally higher diversity. In contrast, in the few-shot setting, MDLM performs slightly better (MDLM: 0.1324 vs GPT-2: 0.1394), suggesting a modest improvement in output diversity compared to GPT-2.

Analysis of Zero-Shot vs Few-Shot Performance Patterns This analysis focuses specifically on the BLEU and ROUGE-L metrics. In the zero-shot setting, MDLM consistently outperforms GPT-2 on both metrics. However, in the few-shot setting, GPT-2 generally achieves better results. This behavior may be explained by the fact that GPT-2 benefits more effectively from in-context examples, which enhances its ability to reproduce surface-level patterns and reference-like phrasing, thereby improving n -gram overlap and subsequence matching. In contrast, MDLM appears to be more robust in zero-shot scenarios, likely due to its training objective, which promotes stronger generalization without relying on explicit demonstrations.

6.2. Manual Analysis

To complement the quantitative metrics, we conducted a human evaluation on a sample of 50 generated sentences. Each prediction was assigned a fidelity score: **0** (hallucination or off-topic), **1** (partially faithful: some facts are correct, others are missing or distorted), and **2** (fully faithful). This qualitative assessment reveals that while neither model is production-ready, they exhibit diametrically opposed strengths and weaknesses.

Architecture-Specific Error Patterns MDLM proves to be fundamentally superior in terms of factual grounding because it actively “reads” the table. When errors occur, they remain anchored to the correct entities, dates, and names. Its failures are primarily formal, such as mechanical repetitions and over-generation, rather than semantic. Simple post-processing, such as truncating at the first period or removing duplicate tokens, would yield usable results in many cases.

In contrast, GPT-2 suffers from a far more critical issue: it produces syntactically fluent sentences that are entirely disconnected from the input data (hallucinations). In a fact-based task, a well-formed but false sentence is more damaging than a poorly formed but true one. Furthermore, this lack of conditioning cannot be corrected by post-processing, as the model fails to learn the relationship between the structured table and the generated output.

cZero-Shot vs. Few-Shot Dynamics The impact of prompting highlights a major architectural divergence. For GPT-2, the transition to a few-shot setting is vital: completely lost in zero-shot, it relies on the provided examples to mimic the expected syntactic structure. While this artificially inflates its ROUGE and BLEU scores, it does not solve the underlying hallucination problem.

Conversely, MDLM excels in zero-shot but is penalized by few-shot prompting. Because its diffusion architecture processes the entire context bidirectionally, unrelated demonstration examples act as "cognitive noise." This irrelevant context distracts the model from the target table, causing a drop in factual precision.

6.3. Full Fine-Tuning on MDLM

Experimental Setup and Rationale To further adapt the pretrained MDLM to the ToTTo domain and assess its potential for performance improvement, we conducted a full fine-tuning experiment. Unlike Parameter-Efficient Fine-Tuning (PEFT) methods, this approach updates all model parameters to maximize domain adaptation. The model was trained for 5 epochs with a learning rate of $5e-05$ and a batch size of 4 on a restricted dataset of 50 training examples. The training process showed rapid convergence, with the continuous-time loss decreasing sharply from an initial value of 1.9759 to a final value of 0.0052. Although this result indicates successful optimization on the training data, the subsequent evaluation on the test set revealed important limitations in the model's generalization ability.

Comparative Analysis of Pre-Trained vs. Fine-Tuned Models Contrary to the initial hypothesis that domain-specific fine-tuning would inherently improve generation quality, the empirical results demonstrate a severe degradation across all evaluated metrics

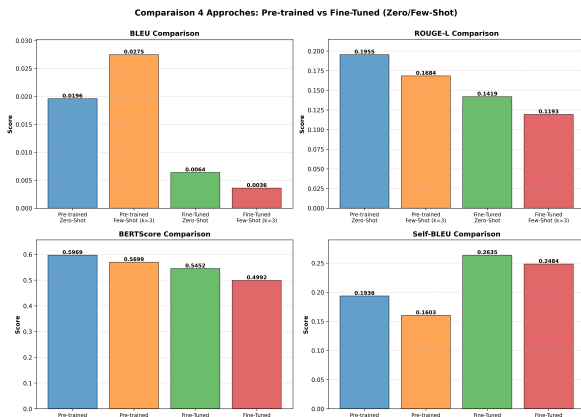


Figure 3: Metrics evaluation : Pre-Trained vs. Fine-Tuned Models

- **Lexical Alignment (BLEU ROUGE-L):** The fine-tuning process caused a precipitous drop in lexical overlap. In the zero-shot setting, the BLEU score collapsed from 0.0196 (pre-trained) to 0.0064 (fine-tuned). The degradation is even more pronounced

in the few-shot ($k=3$) setting, where the BLEU score fell from 0.0275 to an abysmal 0.0036. Similarly, ROUGE-L scores declined from 0.1955 to 0.1419 in zero-shot, and from 0.1684 to 0.1193 in few-shot configurations

- **Semantic Similarity (BERTScore):** The native semantic understanding of the MDLM, which was its strongest asset, was notably impaired. The zero-shot BERTScore dropped from 0.5969 to 0.5452, and the few-shot score dropped from 0.5699 to 0.4992.
- **Generation Diversity (Self-BLEU):** The fine-tuned model exhibited an increase in Self-BLEU scores (e.g., from 0.1936 to 0.2635 in zero-shot), indicating a significant loss of output diversity and a tendency toward repetitive generation.

Statistical Significance and Distribution Analysis To rigorously quantify the impact of fine-tuning, paired t-tests were conducted comparing the BLEU scores of the pre-trained and fine-tuned models. The statistical analysis confirms that the performance deterioration is highly significant. In the zero-shot configuration, the t-statistic is -2.5443 (p-value = 0.014153, thus significant at $\alpha;0.05$). In the few-shot configuration, the degradation is even more robust statistically, yielding a t-statistic of -3.2083 (p-value = 0.002355). The visual analysis of the BLEU score distributions further illustrates this phenomenon. The histograms display a massive shift toward zero for the fine-tuned model distributions. Quantitatively, this represents a relative performance drop of -67.36% in the zero-shot setting and an alarming -87.00% drop in the few-shot setting.

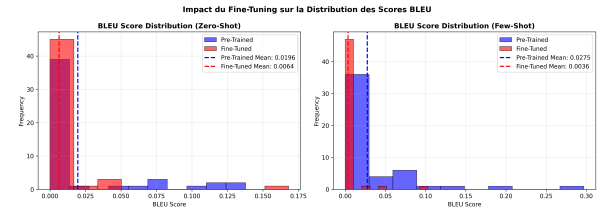


Figure 4: MDLM fine-tuning impact on the BLEU Scores distribution

7. Conclusion

In this study, we investigated the potential of Masked Diffusion Language Models (MDLMs) for structured-to-text generation, comparing a pretrained MDLM against a standard autoregressive baseline (GPT-2) on the ToTTo benchmark. Through a comprehensive evaluation encompassing semantic, lexical, and diversity metrics, alongside qualitative manual analysis, we identified fundamentally different architectural behaviors between sequential prediction and bidirectional denoising.

Our findings demonstrate that autoregressive models like GPT-2 are highly dependent on in-context demonstration. In zero-shot settings, GPT-2 struggled to formulate coherent descriptions from linearized tables and generated severe

hallucinations entirely disconnected from the input data. However, the inclusion of few-shot examples led to drastic improvements in lexical metrics (BLEU and ROUGE-L), confirming its strong capacity to mimic surface-level patterns and phrasing when explicitly guided.

Conversely, the MDLM exhibited a robust "native" understanding of tabular structures. Without any demonstration examples, it achieved the highest overall semantic similarity (BERTScore of 0.5969) and demonstrated superior factual grounding, as its bidirectional attention mechanism actively "reads" the table. Interestingly, introducing few-shot examples to the MDLM caused a degradation in precision; the unrelated demonstration examples acted as "cognitive noise" that distracted the model's bidirectional context window.

Crucially, our investigation into domain adaptation through full fine-tuning highlighted significant vulnerabilities in current diffusion models when operating on limited data. Updating all 0.6 billion parameters on a restricted dataset led to rapid mathematical convergence but resulted in severe overfitting and catastrophic forgetting. The model entirely lost its foundational zero-shot generalization capabilities, leading to a statistically significant collapse across all metrics, including an alarming 87.00% relative drop in few-shot BLEU scores.

Ultimately, this comparative analysis highlights that diffusion-based language models offer a highly promising alternative for non-linear input tasks due to their exceptional zero-shot relational semantics and factual anchoring. However, to properly leverage these architectures for domain-specific tasks without destroying their base capabilities, future work must strictly avoid full fine-tuning on small datasets. Instead, research should prioritize Parameter-Efficient Fine-Tuning (PEFT) strategies and alternative context-integration methods, effectively bridging the gap between deep semantic understanding and exact lexical realization.

References

- Austin et al. (2021). *Structured Denoising Diffusion Models in Discrete State-Spaces*. NeurIPS.
- Ho et al. (2020). *Denoising Diffusion Probabilistic Models*. NeurIPS.
- Lin, C.-Y. (2004). *ROUGE: A Package for Automatic Evaluation of Summaries*. ACL Workshop.
- Lou et al. (2023). *Discrete Diffusion Modeling by Estimating the Ratios of the Data Distribution*. arXiv.
- Ni et al. (2025). *Diffusion Language Models are Super Data Learners*. arXiv.
- Papineni et al. (2002). *BLEU: A Method for Automatic Evaluation of Machine Translation*. ACL.
- Parikh et al. (2020). *ToTTo: A Controlled Table-To-Text Generation Dataset*. EMNLP.
- Radford et al. (2019). *Language Models are Unsupervised Multi-task Learners*. OpenAI Blog.
- Zhang et al. (2020). *BERTScore: Evaluating Text Generation with BERT*. ICLR.